Day 11: Live Coding Session

Bubble sort

Sorting Number → Arrange
- Ascending (ऐसे ही)
- Descending (उल्टे ही)

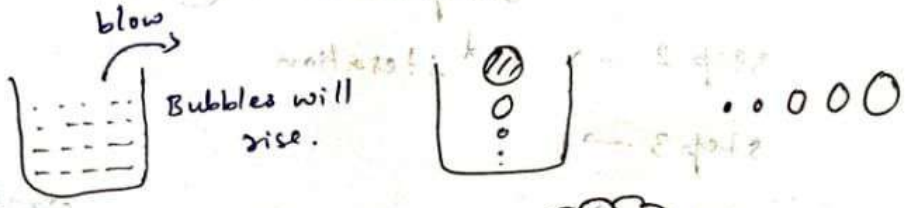
Sorting → letters → J K M Z → Alphabetic position (Comparison)

Letters posn → no,'s

Z J K M

↓

Bubble Sort: There is direct analogy → water Bubbles



Logic:

Array

5	4	19	2	3
0	1	2	3	4

Indices

↑ start a[5] ↑ end (n-1)

⇒ Size = 5 (Total elements)

Sorted

2	3	4	5	19

Step 1>:

5 4 19 2 3

4 5 19 2 3

4 5 19 2 3

4 5 2 19 3

4 5 2 3 19 → biggest is at the end.

→ step 1st result.

1st with 2nd

2nd with 3rd

3rd with 4th

4th with 5th

observe

1st num > 2nd no,

→ 5 > 4 → yes

Swap: 4 place 5

5 place 4

→ 5 > 19 → NO

No action

→ 19 > 2 → yes

swap: 19 → 2

2 → 19

→ 19 > 3 → yes

swap: 19 → 3

3 → 19

Step 2>

4 5 2 3 19

Take result of 1st step and do sorting

→ 4>5
No action

→ 5>2 → yes
Swap: 5→2
2→5

→ 5>3 → yes
Swap: 5→3
3→5

→ 5>19
no action

4 5 2 3 19

4 5 2 3 19

4 2 5 3 19

4 2 3 5 19

→ 2nd step result.

Time Com

1 step	5 steps
5 times	25

5×5
$O(n \times n)$
$O(n^2)$

step 1 → iteration → 1
Comparion → 5 time

step 2 → 2nd iteration

step 3 →

step 4 →

step 5 →

space → memory
↓
variables

1 loop for Traversing
2 loop for Comparision

loop ಒಳಗೆ loop
for(iteration)
for(comparion)

Step 3>

4 2 3 5 19

2 4 3 5 19

2 3 4 5 19 ×

2 3 4 5 19

→ 4>2
Swap: 4→2
2→4

→ 4>3
Swap: 4→3
3→4

→ 4>5
No action

~~Step 4>~~

↳ Decending

4 5 3 2 1

worst case 5×5

1 2 3 4 5

↳ Best case

1 2 3 5 4

Requirement :

1. 2 loops
2. $1^{st} > 2^{nd}$ → if ( )
3. Swapping

0	1	2	3	4
5	4	19	2	3

5 > 4 → yes, swap

left right

temp = a[0]

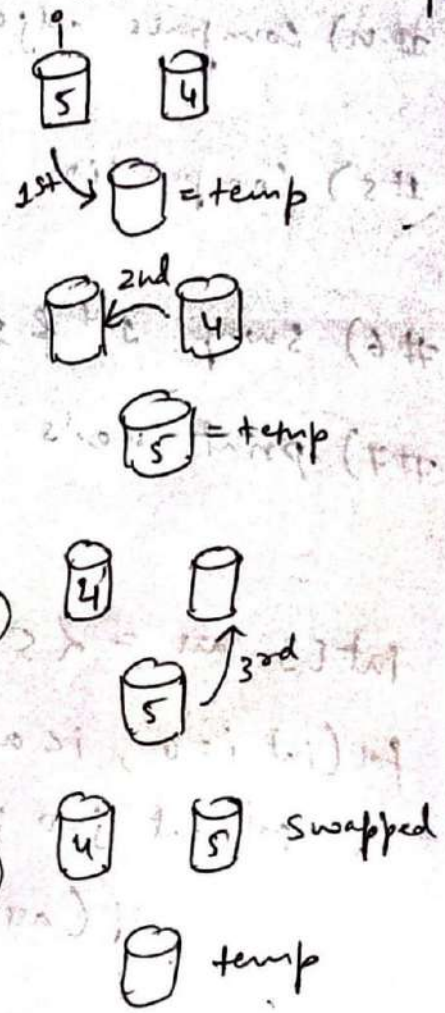
iteration

```
for (i=0; i<n-1; i++)
    for (j=0; j<n-1; j++)
        if (a[0] > a[1])
        {
            swap
        }
return no,
```

Comparision

(or) Print

j > j+1



Step 1 : get input as array

Step 2 : Compare the adjacent no's

#1) Take a list of no's

#2) Find size(n)

#3) Traverse three array → outer loop start = 0 end = n-1

#4) compare adjacent elements → we need inner loop
start = 0 end = n-1

#5) Comparision if (1st > 2nd) (2nd > 3rd) } yes

#6) swap 1st & 2nd, 2nd & 3rd so on

#7) print no's — sorted

```
int[] arr = {5, 4, 19, 2, 3};
for (int i=0; i< arr.length -1; i++) {
    for (int j=0; j< arr.length -i-1; j++) {
        if (arr[j] > arr[j+1]) {
            int temp = arr[j];
            arr[j] = arr[j+1];
            arr[j+1] = temp;
        }
    }
}
for (int i=0; i<arr.length; i++) {
    System.out.print(arr[i] + " ");
}
```